

Администрирование локальных сетей

Лабораторная работа №2

Абдуллахи Бахара

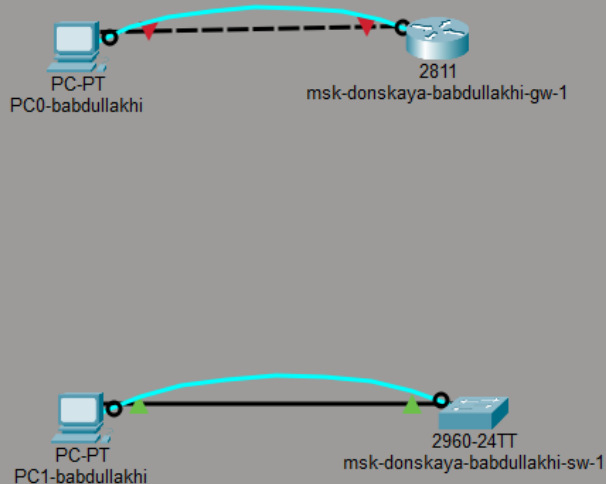
16 февраля 2026

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Получить основные навыки по начальному конфигурированию сетевого оборудования Cisco.

Выполнение лабораторной работы



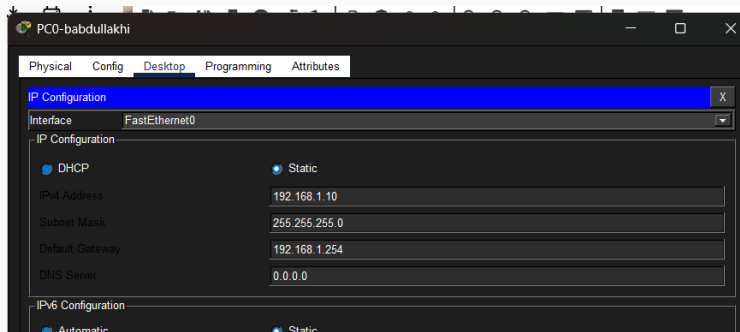
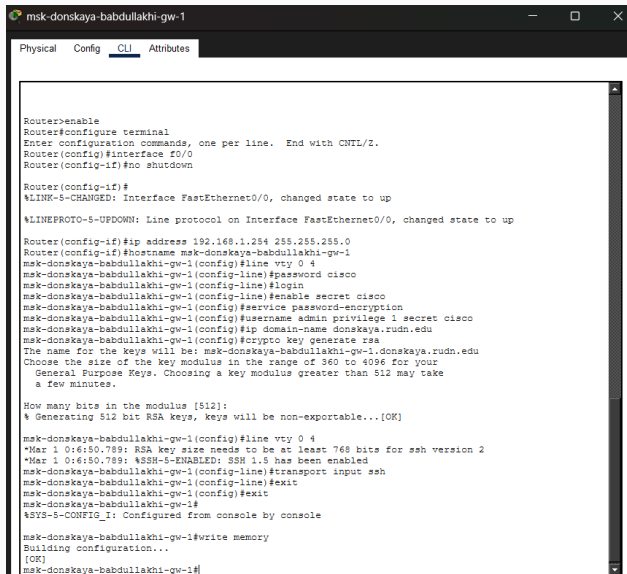


Рис. 2: Настройка IP-адреса на PC0

Настройка маршрутизатора



```
msk-donskaya-babdullakhi-gw-1
Physical Config CLI Attributes

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

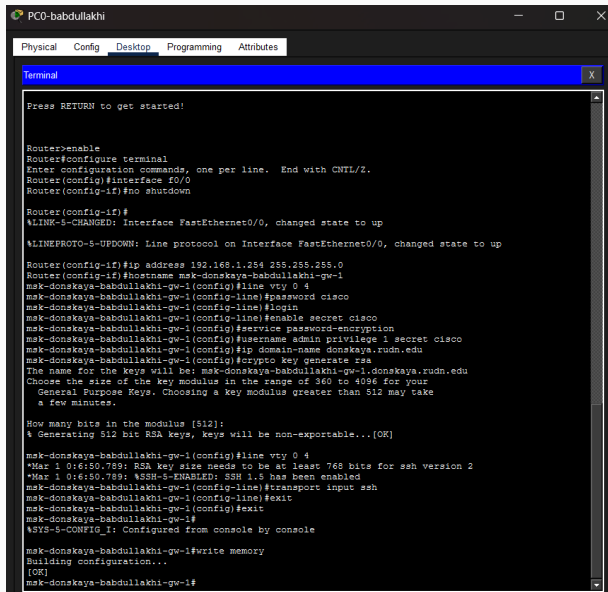
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#hostname msk-donskaya-babdullakhi-gw-1
msk-donskaya-babdullakhi-gw-1(config)#line vty 0 4
msk-donskaya-babdullakhi-gw-1(config-line)#password cisco
msk-donskaya-babdullakhi-gw-1(config-line)#login
msk-donskaya-babdullakhi-gw-1(config-line)#enable secret cisco
msk-donskaya-babdullakhi-gw-1(config)#service password-encryption
msk-donskaya-babdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-babdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-babdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:6:50.789: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:50.789: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-babdullakhi-gw-1(config-line)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#write memory
Building configuration...
[OK]
msk-donskaya-babdullakhi-gw-1#
```

Настройка маршрутизатора через консоль



```
PC0-babdullakhi
Physical Config Desktop Programming Attributes
Terminal
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#hostname msk-donskaya-babdullakhi-gw-1
msk-donskaya-babdullakhi-gw-1(config)#line vty 0 4
msk-donskaya-babdullakhi-gw-1(config-line)#password cisco
msk-donskaya-babdullakhi-gw-1(config-line)#login
msk-donskaya-babdullakhi-gw-1(config-line)#enable secret cisco
msk-donskaya-babdullakhi-gw-1(config)#service password-encryption
msk-donskaya-babdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-babdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-babdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:6:50.789: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:50.789: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-babdullakhi-gw-1(config-line)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#write memory
Building configuration...
[OK]
msk-donskaya-babdullakhi-gw-1#
```

Проверка подключения к маршрутизатору

```
PC0-babdullakhi
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time=4ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>telnet 192.168.1.254
Trying 192.168.1.254 ...Open

[Connection to 192.168.1.254 closed by foreign host]
C:\>ssh -l admin 192.168.1.254

Password:

msk-donskaya-babdullakhi-gw-1>
msk-donskaya-babdullakhi-gw-1>
msk-donskaya-babdullakhi-gw-1>exit

[Connection to 192.168.1.254 closed by foreign host]
C:\>
```

Настройка коммутатора

```
msk-donskaya-babdullakhi-sw-1
Physical Config CLI Attributes
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-babdullakhi-sw-1
msk-donskaya-babdullakhi-sw-1(config)#interface vlan2
msk-donskaya-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#no shutdown
msk-donskaya-babdullakhi-sw-1(config-if)#ip address 192.168.2.1 255.255.255.0
msk-donskaya-babdullakhi-sw-1(config-if)#interface f0/1
msk-donskaya-babdullakhi-sw-1(config-if)#switchport mode access
msk-donskaya-babdullakhi-sw-1(config-if)#switchport access vlan 2
msk-donskaya-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#ip default-gateway 192.168.2.254
msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4
msk-donskaya-babdullakhi-sw-1(config-line)#password cisco
msk-donskaya-babdullakhi-sw-1(config-line)#login
msk-donskaya-babdullakhi-sw-1(config-line)#line console 0
msk-donskaya-babdullakhi-sw-1(config-line)#password cisco
msk-donskaya-babdullakhi-sw-1(config-line)#login
msk-donskaya-babdullakhi-sw-1(config-line)#enable secret cisco
msk-donskaya-babdullakhi-sw-1(config)#service password-encryption
msk-donskaya-babdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-babdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-babdullakhi-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:14:13.440: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:14:13.440: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-sw-1(config-line)#transport input ssh
msk-donskaya-babdullakhi-sw-1(config-line)#exit
msk-donskaya-babdullakhi-sw-1(config)#exit
msk-donskaya-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

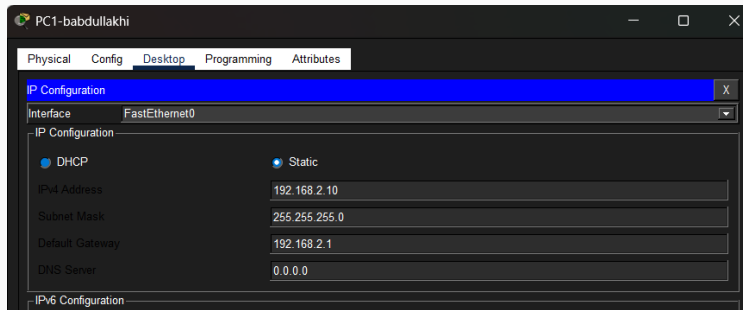
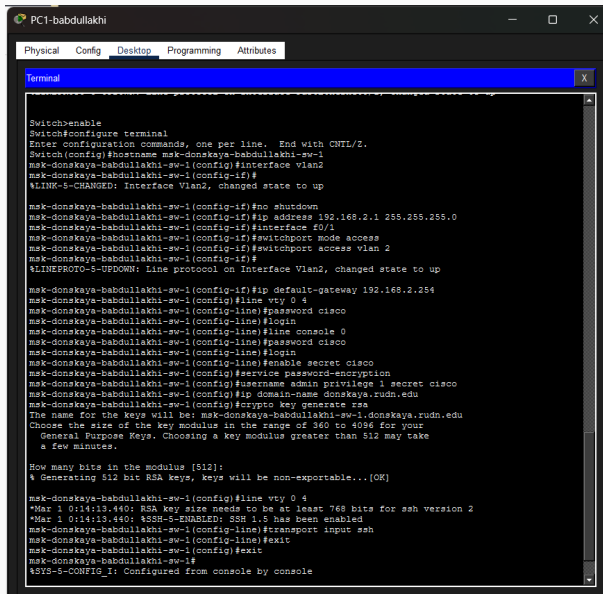


Рис. 7: Настройка IP-адреса на PC1

Настройка коммутатора через консоль



```
PC1-babdullakhi
Physical Config Desktop Programming Attributes
Terminal

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-babdullakhi-sw-1
msk-donskaya-babdullakhi-sw-1(config)#interface vian2
msk-donskaya-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vian2, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#no shutdown
msk-donskaya-babdullakhi-sw-1(config-if)#ip address 192.168.2.1 255.255.255.0
msk-donskaya-babdullakhi-sw-1(config-if)#interface f0/1
msk-donskaya-babdullakhi-sw-1(config-if)#switchport mode access
msk-donskaya-babdullakhi-sw-1(config-if)#switchport access vlan 2
msk-donskaya-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#ip default-gateway 192.168.2.254
msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4
msk-donskaya-babdullakhi-sw-1(config-line)#password cisco
msk-donskaya-babdullakhi-sw-1(config-line)#login
msk-donskaya-babdullakhi-sw-1(config-line)#line console 0
msk-donskaya-babdullakhi-sw-1(config-line)#password cisco
msk-donskaya-babdullakhi-sw-1(config-line)#login
msk-donskaya-babdullakhi-sw-1(config-line)#enable secret cisco
msk-donskaya-babdullakhi-sw-1(config)#service password-encryption
msk-donskaya-babdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-babdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-babdullakhi-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:14:13.440: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:14:13.440: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-sw-1(config-line)#transport input ssh
msk-donskaya-babdullakhi-sw-1(config-line)#exit
msk-donskaya-babdullakhi-sw-1(config)#exit
msk-donskaya-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Проверка подключения к коммутатору

```
PC1-babdullakhi
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.2.1
Trying 192.168.2.1 ...Open

[Connection to 192.168.2.1 closed by foreign host]
C:\>ssh -l admin 192.168.2.1

Password:

msk-donskaya-babdullakhi-sw-1>
msk-donskaya-babdullakhi-sw-1>
msk-donskaya-babdullakhi-sw-1>exit

[Connection to 192.168.2.1 closed by foreign host]
C:\>
```

Выводы по проделанной работе

В ходе лабораторной работы была собрана сеть в Cisco Packet Tracer, выполнена базовая настройка маршрутизатора и коммутатора, назначены IP-адреса интерфейсам и настроен доступ по паролю и SSH. Проверка с помощью команды ping подтвердила корректную работу соединений и доступность сетевых устройств.